Discovering the Signal: It Lives in the Tail

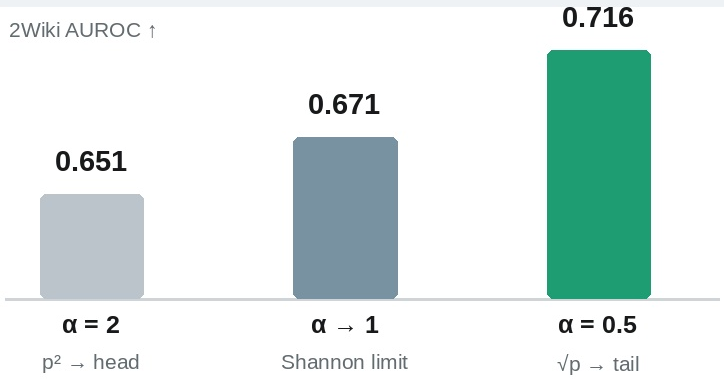
42-signal survey → entropy decomposition

1 Broad survey

42 confidence signals · 4 datasets × 1,000 questions

$$H_\alpha(p) = \frac{1}{(1-\alpha)} \cdot \log \sum_v p(v)^\alpha$$

α > 1: head-heavy · α → 1: Shannon · α < 1: tail-sensitive



Lower α gives rare tokens more influence → better separation

2 Decompose entropy

A = top-5, B = tail, $m_5 = P(A)$

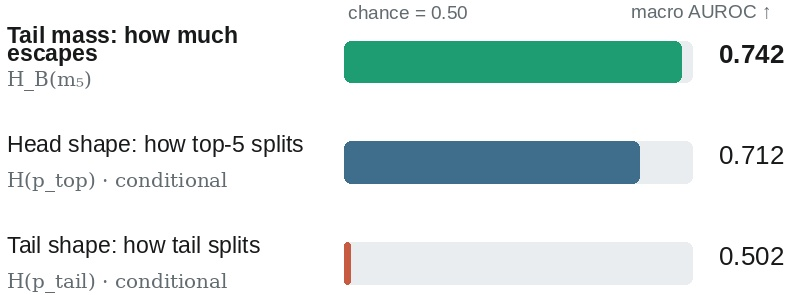
$p_{top}(v) = p(v)/m_5$ on A; $p_{tail}(v) = p(v)/(1-m_5)$ on B

$H(p) = -\sum_{v \in A} p(v) \log p(v) - \sum_{v \in B} p(v) \log p(v)$

Substitute $p = m_5 p_{top}$ on A and $p = (1-m_5) p_{tail}$ on B; expand log

$$\Rightarrow H = H_B(m_5) + m_5 H(p_{top}) + (1-m_5) H(p_{tail})$$

Diagnostic probes isolate shape: H(p_tail) normalizes away tail amount



Amount matters (.742); allocation inside the tail does not (.502)

Takeaway: tail-sensitive signals detect how much mass escapes the head—not how that tail mass is arranged.

From Entropy to tl_sum

isolate the escape signal → make it monotonic → aggregate across tokens

1 Identify the strongest term

$$H_B(m_5)$$

escape 0.742 > full entropy 0.726

discard within-top-5 and tail-composition terms

→

2 Design a monotonic score

$$H_B(m_5) \rightarrow -\log m_5^{(t)}$$

H_B is not globally monotonic

less top-5 mass → larger score

→

3 Aggregate over the answer

$$tl_sum = \sum_t -\log m_5^{(t)} = -\log \prod_t m_5^{(t)}$$

joint surprisal of staying inside top-5

Why sum?

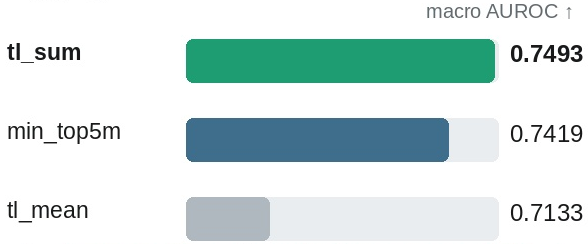
wrong answers often leak a little at several positions

token t_1 — small leak + token t_2 — small leak + token t_3 — small leak → Σ tl_sum

Worst token: keeps one position

tl_sum: all positions accumulate

Aggregation scan



tl_sum vs. min_top5m: paired bootstrap p < 0.001

Takeaway: tl_sum is built from the top-5 escape signal identified by entropy decomposition—not an ad hoc formula.